Stakeholder Analysis

Project: Real-Time IoT Data Pipeline **Team Leader:** Rana Shaaban Abdelqader Mohamed
Section: 3. Requirements Gathering

1. Purpose

This document identifies everyone who is impacted by or invested in the project, records what each stakeholder needs, and shows how much influence and interest each one has. It is the starting point for the user stories, functional requirements, and non-functional requirements that follow.

2. Stakeholder Register

ID	Stakeholder	Type	Role in the Project
S1	Lecturer	Internal (academic)	Defines the course requirements, reviews weekly deliverables, and grades the final result
S2	Project team (5 members)	Internal	Designs, builds, tests, and documents the pipeline
S3	Dashboard users (monitoring operators)	External (end users)	Watch live sensor trends and respond to alerts
S4	Technology providers (Apache Kafka, Snowflake, Streamlit, GitHub)	External	Supply the tools and platforms the project depends on (open source or free/trial tiers)

3. Stakeholder Needs and Expectations

ID	Stakeholder	Core Needs and Expectations	Success Looks Like
S1	Lecturer	A complete, working project with both a batch and a streaming pipeline, all weekly deliverables, and clear documentation uploaded to GitHub	All four weekly deliverables are submitted on time and the final demo works end to end
S2	Project team	A fair task split, hands-on practice with Kafka, ETL, streaming, and dashboards, and a successful final demo	Each member owns one clear component and the five components integrate without blocking each other
S3	Dashboard users	Immediate alerts when temperature or humidity crosses a threshold, a live view of sensor trends, and accurate, easy-to-read metrics	An alert appears within 5 seconds of a breach and the dashboard is readable without training
S4	Technology providers	Tools are used within their free or open-source terms	Project stays within free tiers and trial limits

4. Influence / Interest Mapping

Stakeholder	Influence	Interest	Strategy
Lecturer	High	High	**Manage closely:** confirm requirements early, share progress every week, and get sign-off on the requirements baseline
Project team	High	High	**Manage closely:** agree interfaces and the data schema in Week 1, hold regular sync meetings
Dashboard users	Low	Medium	**Keep informed:** validate dashboard layout and alert wording with a wireframe before building

Stakeholder	Influence	Interest	Strategy
Technology providers	Low	Low	**Monitor:** check free-tier and trial limits at the start of each phase

5. Elicitation Techniques per Stakeholder

Stakeholder	Technique	What We Want to Learn
Lecturer	Review of the course brief and grading criteria, plus a short requirements walkthrough	Mandatory deliverables, grading weights, documentation format
Project team	Collaborative workshop	Component boundaries, interfaces, schema, and task ownership
Dashboard users	Short survey or interview with a sample operator, plus a dashboard wireframe review	Which metrics matter, alert wording, and refresh expectations
Technology providers	Documentation review	Limits, setup requirements, and fallback options

6. Potential Conflicts and Alignment Risks

#	Potential Conflict	Between	Early Resolution
1	The lecturer expects a complete batch and streaming pipeline, but the 4-week timeline is tight for a team new to Kafka and Snowflake	S1 and S2	Freeze scope in Week 1, share tutorials, pair members on first tasks
2	Operators want fast, frequent updates, but the batch path loads data to Snowflake with a delay	S3 and S2	Show alerts from the streaming path for speed and processed trends from Snowflake for accuracy

#	Potential Conflict	Between	Early Resolution
3	Schema changes by one member can break another member's component	S2 internally	Freeze the schema in Week 1; changes need team leader approval

7. Engagement Plan

Stakeholder	When	How
Lecturer	Weekly, and at each deliverable	Deliverable submission, short progress update, requirements sign-off
Project team	Continuous	Team chat, weekly sync, GitHub issues and pull requests
Dashboard users	Before and after the dashboard build (Week 4)	Wireframe review, then demo feedback
Technology providers	Start of each phase	Documentation and limits check